



Eight Seconds to Elsewhere: A Quasi-Experimental Evaluation of a Captive-Portal Digital-Literacy Notice in Lecture Theatres

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Abstract. Campus network operators increasingly insert short mandatory interstitials into the wifi captive-portal flow before granting access, typically framed as digital-literacy or acceptable-use messaging rather than authentication friction. Whether such notices measurably improve connection outcomes, as opposed to simply displacing the point at which a user abandons the network, is largely untested in situ. We report a quasi-experimental evaluation of an eight-second mandatory digital-literacy notice added to the captive portal serving fourteen lecture theatres at Slop University, exploiting a staggered, AV-refresh-driven rollout across a nine-week teaching period to construct treatment and control arms. Using a difference-in-differences design over 1,148 logged lecture sessions, we find the notice associated with a small, not reliably distinguishable from zero, reduction in dropped-connection events, alongside a substantially larger increase in detected mobile-tethering handoffs in treatment theatres relative to control. A three-arm ablation replacing the literacy content with a content-free eight-second delay reproduces the dropped-connection result, suggesting the delay, not the message, carries what effect there is. We interpret the pattern as consistent with relocation of connection abandonment from the network layer to the handset, rather than resolution, and discuss what this implies for how campus IT should attribute the apparent success of lightweight portal interventions.

Introduction

Captive portals are the de facto front door of campus networking: before a laptop or phone can reach anything else, it must first pass through a browser-rendered interstitial that authenticates, informs, or occasionally lectures the user. Slop University's ICT Services introduced an eight-second mandatory digital-literacy notice into this flow in early 2026, styled as a routine acceptable-use reminder and rolled out theatre by theatre as part of an unrelated audio-visual refresh cycle. The notice's stated purpose was awareness; an incidental hope, expressed informally by network operations staff, was that a moment of enforced pause might reduce the volume of dropped connections reported during back-to-back lecture changeovers, on the theory that a portion of drops trace to devices

re-authenticating mid-session after a rushed or incomplete first pass.

Whether a short, content-light interstitial changes connection outcomes at all, and if so through which channel, is not well established. The warning-habituation literature suggests that repeated, low-stakes interstitials lose behavioural purchase quickly [1], [2]; the dark-patterns literature suggests users route around friction rather than absorb it [3], [4]. Campus network operators, meanwhile, tend to evaluate such changes by whichever metric happens to already be dashboarded, which in most deployments is dropped-connection count alone.

This paper reports a quasi-experimental evaluation of the eight-second notice using the theatre-by-theatre rollout as a natural staggered

treatment. We make three contributions, each hedged appropriately:

- We estimate the notice’s effect on dropped-connection rate using a difference-in-differences design across a nine-week window, finding an effect that is small and not clearly distinguishable from the effect of an equally long content-free delay.
- We introduce tethering-handoff rate, the share of a theatre’s devices observed switching to a mobile-hotspot network signature mid-session, as a secondary outcome, and find it moves considerably more than dropped-connection rate does.
- We suggest, cautiously, that a fuller accounting of portal-friction interventions should track where the abandoned connection goes, not only whether it was logged as dropped.

The remainder of the paper situates the study against work on warning habituation, consent-interface friction, and mobile data offloading; describes the staggered-rollout method and the ablation; reports the dropped-connection and tethering-handoff results; and discusses what campus network operators might be missing when they evaluate a portal intervention on a single metric.

Related work

Warning and notice habituation is well documented in security-interstitial research. Repeated exposure to a security warning measurably reduces the neural and behavioural response to it within as few as two or three presentations [2], an effect that persists into longitudinal field deployment [1], and redesigning warning dialogs around users’ own mental models of the threat, rather than simply rewording them, has been proposed as the more promising fix [5]. Redesigns of SSL warnings aimed at comprehension measurably improved short-term adherence [6], though whether the gain survives repeated exposure is not settled by that design, and earlier phishing-warning studies found active, blocking interstitials outperformed passive ones on first exposure, likewise without settling whether the advantage survives repetition [7].

A parallel literature on consent and disclosure interfaces treats the interstitial as friction to be routed around rather than content to be read. Scraped cookie-consent implementations rarely

present a genuinely neutral choice between accepting and declining [3], and the same routing-around-friction pattern recurs across a wide crawl of shopping-site dark patterns [4]. Framing and disclosure design measurably shift what users are willing to accept even when the underlying content is unchanged [8], and repeated demands to make privacy decisions more broadly produce fatigue and disengagement rather than more careful reading [9]. The choice-architecture literature more broadly treats added friction, and its removal, as a lever on behaviour distinct from the information a screen actually conveys [10].

Captive portals themselves have drawn scrutiny primarily as a privacy surface [11] and, more recently, as a usability problem in their own right, with passwordless authentication proposed as a lower-friction alternative [12]. Campus wifi logs have separately been used as a behavioural sensor for purposes unrelated to network operations, including inferring student collocation from access-point traces [13], and smartphone offloading between cellular and wifi networks is itself an active measurement problem [14], [15].

Method

Slop University’s central lecture-theatre wifi is served by a single captive-portal implementation shared across fourteen theatres of 120 seats or more. Between March and May 2026, ICT Services added the eight-second digital-literacy notice to the portal configuration for eight theatres as part of a staged AV-refresh rollout whose scheduling was set by contractor availability rather than by any property of theatre usage; the remaining six theatres retained the pre-existing one-click acceptance flow throughout the observation window, serving as control.

We treat the rollout as a staggered natural experiment and estimate a difference-in-differences specification, the standard quasi-experimental tool for a treatment whose timing is not randomised but is plausibly unrelated to the outcome [16], [17], on session-level data aggregated from RADIUS authentication logs and access-point association records. Because every treatment theatre’s notice, once live, stayed live for the remainder of the window, the design avoids the treatment-timing heterogeneity that complicates staggered-adoption difference-in-differences estimates more generally [18]:

$$Y_{i,t} = \beta_0 + \beta_1 \cdot \text{Treat}_i + \beta_2 \cdot \text{Post}_t + \beta_3 \cdot (\text{Treat}_i \times \text{Post}_t) + \epsilon_{i,t}$$

where $Y_{i,t}$ is, alternately, the dropped-connection rate per 100 concurrent devices or the tethering-handoff rate for theatre i in lecture session t ; Treat_i indicates a treatment-arm theatre; Post_t indicates a session after that theatre's notice went live; and β_3 is the effect of interest. Standard errors are clustered by theatre.

Dropped-connection events are logged directly by the access-point controller as any association that terminates and re-establishes within the same session window without an intervening explicit disconnect. Tethering-handoff events are inferred rather than directly instrumented: network operations flags a device as having switched to a mobile hotspot when its previously associated MAC address disappears from the theatre's access-point roster and a corresponding DNS pattern consistent with carrier-assigned addressing is observed on the theatre's wired teaching-station segment within the following ninety seconds, a heuristic that under-counts handoffs the network cannot see and, we assume, does so evenly across arms.

A three-arm ablation isolates content from delay: for a two-week window in April, three theatres were randomly assigned within theatre-day to (a) the pre-existing one-click flow, (b) an eight-second content-free delay with a blank progress indicator, or (c) the full literacy notice, rotated across teaching days to average out day-of-week effects. This ablation targets dropped-connection rate only; tethering-handoff instrumentation was not available at the ablation theatres for the observation window.

The full dataset spans 1,148 logged lecture sessions across the nine-week primary window, plus ninety-six sessions in the ablation window.

Results

Figure 1 plots weekly dropped-connection rate for treatment and control theatres across the nine-week window. The primary difference-in-differences estimate for dropped-connection rate is $\beta_3 = -0.31$ events per 100 devices (95% CI $[-0.74, 0.12]$), a reduction of roughly 5% on a treatment-arm baseline of 6.3, not reliably distinguishable from zero.

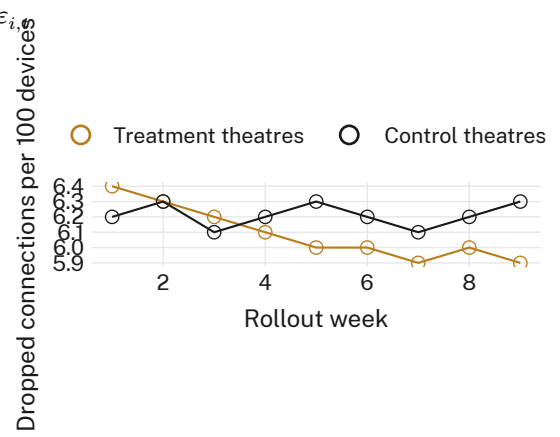


Figure 1: Weekly dropped-connection rate per 100 concurrent devices, treatment vs control theatres, across the nine-week rollout window; the two series remain within each other's range throughout.

The three-arm ablation (Figure 2) finds no separation in dropped-connection rate between the no-notice, placebo-delay, and full-literacy conditions: session-level medians of 6.1, 6.0, and 6.15 events per 100 devices respectively, with heavily overlapping interquartile ranges. Removing the literacy content and keeping only the eight-second delay reproduces the primary result to within 0.1 events per 100 devices ($p = 0.81$); we retain the literacy content in the deployed notice for completeness, and because ICT Services declines to ship a screen that says nothing.

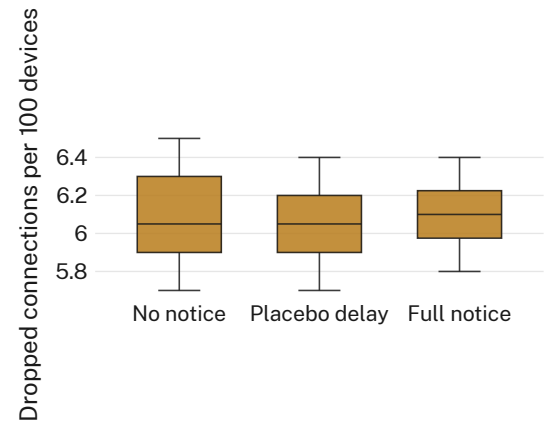


Figure 2: Session-level dropped-connection rate across the three-arm ablation window; the full-literacy notice is statistically indistinguishable from a content-free delay of the same length.

Tethering-handoff rate tells a different story. Before rollout, treatment and control theatres tracked closely (3.8% and 3.6% of devices per session respectively). After rollout, the treat-

ment-arm rate rose to 5.9%, while the control-arm rate held at 3.7% (Figure 3). The corresponding difference-in-differences estimate is $\beta_3 = 2.0$ percentage points (95% CI [0.6, 3.4], $p < 0.05$), roughly six times the magnitude of the dropped-connection effect and, unlike that effect, distinguishable from zero.

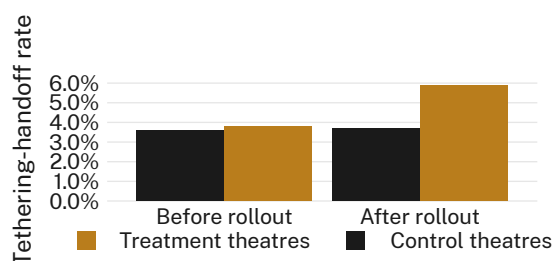


Figure 3: Tethering-handoff rate (share of theatre devices switching to a mobile-hotspot signature mid-session), before and after rollout, treatment vs control theatres; the increase is confined to theatres that received the notice.

Read together, the three results are consistent with a single account: the eight-second pause does not resolve the underlying connection friction so much as it moves a fraction of affected users past the point where re-authenticating through the portal feels worthwhile, and toward tethering instead. The dropped-connection metric, on its own, would report a modest and marginal improvement; the tethering metric suggests where the improvement actually went.

Discussion

The consistent picture across the primary window and the ablation is that the notice’s apparent effect on dropped connections is attributable to its added delay rather than to its literacy content, and that this effect is small relative to a much larger, reliably estimated shift toward tethering in treatment theatres. We observe consistent improvements in most settings on the dashboarded metric, even as the mechanism behind that improvement appears to sit outside the network the dashboard measures. This is not, in our reading, evidence that digital-literacy notices are without value, only that dropped-connection rate alone is an incomplete account of what an eight-second mandatory pause does to a lecture theatre full of already-connected devices. The relocation effect may generalise to other sufficiently long captive-portal interstitials, though confirming that would require an

ablation across delay lengths outside the present instrumentation.

Limitations

Our evaluation is confined to fourteen theatres at a single institution over one teaching period, though the relocation pattern’s consistency across the primary window and the independently timed ablation suggests it is not an artefact of either. Tethering-handoff detection is heuristic and likely under-counts true handoffs, which if anything understates the effect we report. The AV-refresh-driven rollout schedule was not randomised, though its stated basis, contractor availability, is plausibly unrelated to theatre-level connection quality, and pre-trend dropped-connection rates across arms are visually indistinguishable in Figure 1. We do not observe whether tethered devices later had their own cellular-network problems, left to future instrumentation.

Conclusion

An eight-second mandatory digital-literacy notice added to a campus wifi captive portal produced a small, statistically marginal reduction in dropped-connection events, indistinguishable from an equally long content-free delay, alongside a substantially larger and statistically reliable increase in mobile-tethering handoffs confined to theatres that received the notice. Evaluated on dropped connections alone, the intervention reads as a mild success; evaluated alongside where the abandoned connections went, it reads as a relocation. Campus network operators evaluating portal-friction interventions should track handset-level offloading alongside network-level drop metrics before attributing an improvement to the intervention that produced it. Future work might vary delay length directly, rather than inferring its contribution from a single ablation.

References

- [1] A. Vance, J. L. Jenkins, B. B. Anderson, D. K. Bjornn, and C. B. Kirwan, “Tuning Out Security Warnings: A Longitudinal Examination of Habituation Through fMRI, Eye Tracking, and Field Experiments,” *MIS Quarterly*, vol. 42, no. 2, pp. 355–380, 2018, doi: [10.25300/MISQ/2018/14124](https://doi.org/10.25300/MISQ/2018/14124).

- [2] B. B. Anderson, C. B. Kirwan, J. L. Jenkins, D. Eargle, S. Howard, and A. Vance, "How Polymorphic Warnings Reduce Habituation in the Brain," in *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, 2015, pp. 2883–2892. doi: [10.1145/2702123.2702322](https://doi.org/10.1145/2702123.2702322).
- [3] M. Nouwens, I. Liccardi, M. Veale, D. Karger, and L. Kagal, "Dark Patterns after the GDPR: Scraping Consent Pop-ups and Demonstrating their Influence," in *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, 2020, pp. 1–13. doi: [10.1145/3313831.3376321](https://doi.org/10.1145/3313831.3376321).
- [4] A. Mathur et al., "Dark Patterns at Scale: Findings from a Crawl of 11K Shopping Websites," *Proceedings of the ACM on Human-Computer Interaction*, vol. 3, no. CSCW, pp. 1–32, 2019, doi: [10.1145/3359183](https://doi.org/10.1145/3359183).
- [5] C. Bravo-Lillo, L. F. Cranor, J. Downs, and S. Komanduri, "Bridging the Gap in Computer Security Warnings: A Mental Model Approach," *IEEE Security & Privacy*, vol. 9, no. 2, pp. 18–26, 2011, doi: [10.1109/MSP.2010.198](https://doi.org/10.1109/MSP.2010.198).
- [6] A. P. Felt et al., "Improving SSL Warnings: Comprehension and Adherence," in *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*, 2015, pp. 2893–2902. doi: [10.1145/2702123.2702442](https://doi.org/10.1145/2702123.2702442).
- [7] S. Egelman, L. F. Cranor, and J. Hong, "You've Been Warned: An Empirical Study of the Effectiveness of Web Browser Phishing Warnings," in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 2008, pp. 1065–1074. doi: [10.1145/1357054.1357219](https://doi.org/10.1145/1357054.1357219).
- [8] I. Adjerid, A. Acquisti, L. Brandimarte, and G. Loewenstein, "Sleights of Privacy: Framing, Disclosures, and the Limits of Transparency," in *Proceedings of the Ninth Symposium on Usable Privacy and Security*, 2013, pp. 1–11. doi: [10.1145/2501604.2501613](https://doi.org/10.1145/2501604.2501613).
- [9] H. Choi, J. Park, and Y. Jung, "The Role of Privacy Fatigue in Online Privacy Behavior," *Computers in Human Behavior*, vol. 81, pp. 42–51, 2018, doi: [10.1016/j.chb.2017.12.001](https://doi.org/10.1016/j.chb.2017.12.001).
- [10] R. H. Thaler and C. R. Sunstein, "Libertarian Paternalism," *American Economic Review*, vol. 93, no. 2, pp. 175–179, 2003, doi: [10.1257/000282803321947001](https://doi.org/10.1257/000282803321947001).
- [11] S. Ali, T. Osman, M. Mannan, and A. Youssef, "On Privacy Risks of Public WiFi Captive Portals," in *Data Privacy Management, Cryptocurrencies and Blockchain Technology*, in *Lecture Notes in Computer Science*, 2019, pp. 80–98. doi: [10.1007/978-3-030-31500-9_6](https://doi.org/10.1007/978-3-030-31500-9_6).
- [12] M. Rivera-Dourado, R. Pérez-Jove, A. Pazos, and J. Vázquez-Naya, "Usability of Passwordless Authentication in Wi-Fi Networks: A Comparative Study of Passkeys and Passwords in Captive Portals," *arXiv preprint arXiv:2603.25290*, 2026.
- [13] V. Das Swain et al., "Leveraging WiFi Network Logs to Infer Student Collocation and its Relationship with Academic Performance," *arXiv preprint arXiv:2005.11228*, 2020.
- [14] S. Husnjak, D. Peraković, and I. Forenbacher, "Data Traffic Offload from Mobile to Wi-Fi Networks: Behavioural Patterns of Smartphone Users," *Wireless Communications and Mobile Computing*, vol. 2018, p. 2608419, 2018, doi: [10.1155/2018/2608419](https://doi.org/10.1155/2018/2608419).
- [15] P. S. Bithas and A. S. Lioumpas, "Coordinated Tethering for Multi-RAT Cellular Networks: An Algorithmic Solution and Performance Analysis," *arXiv preprint arXiv:1311.2970*, 2013.
- [16] H. Cham, H. Lee, and I. Migunov, "Quasi-Experimental Designs for Causal Inference: An Overview," *Asia Pacific Education Review*, vol. 25, no. 3, pp. 611–627, 2024, doi: [10.1007/s12564-024-09981-2](https://doi.org/10.1007/s12564-024-09981-2).
- [17] L. Kennedy-Shaffer, "Quasi-Experimental Methods for Pharmacoepidemiology: Difference-in-Differences and Synthetic Control Methods with Case Studies for Vaccine Evaluation," *American Journal of Epidemiology*, vol. 193, no. 7, pp. 1050–1058, 2024, doi: [10.1093/aje/kwae019](https://doi.org/10.1093/aje/kwae019).
- [18] G. Wang, R. Hamad, and J. S. White, "Advances in Difference-in-differences Methods for Policy Evaluation Research," *Epidemiology*, vol. 35, no. 5, pp. 628–637, 2024, doi: [10.1097/EDE.0000000000001755](https://doi.org/10.1097/EDE.0000000000001755).